Third Problem Sheet
on
Complex Analysis
Autumn 2026

Tutor: Ronny Morales Carmona
Topics: Infinite Series
Power Series
Sequences of Functions

"A mathematician, like a painter or a poet, is a maker of patterns. If his patterns are more permanent than theirs, it is because they are made with ideas."
-G.H. Hardy

Problem 1

Use mathematical induction to prove that

$$\sum_{k=0}^{n} z^k = \frac{1 - z^{n+1}}{1 - z} \quad \text{for all } z \in \mathbb{C} \setminus \{1\}.$$

Problem 2

Prove Theorem 4.3 (Properties of Series).

Problem 3

Show that if $X = (x_n)$ and $Y = (y_n)$ are real sequences and if for some $K \in \mathbb{N}$ we have

$$0 \leq x_n \leq y_n \quad \text{for} \quad n \geq K$$

then:

i) the convergence of $\sum y_n$ implies the convergence of $\sum x_n$.
ii) the divergence of $\sum x_n$ implies the divergence of $\sum y_n$.

Problem 4

If $\lim(x_n) = x > 0$, show that there exists a natural number K such that if $n \geq K$, then $\frac{1}{2}x < x_n < 2x$. Use this result to prove the **Limit comparison test**: If $X = (x_n)$ and $Y = (y_n)$ are strictly positive sequences and if

$$r := \lim\left(\frac{x_n}{y_n}\right)$$

exists in $\mathbb{R}$, then

i) if $r \neq 0$, then $\sum x_n$ is convergent if and only if $\sum y_n$ is convergent.

ii) If $r = 0$ and if $\sum y_n$ is convergent, then $\sum x_n$ is convergent.

Problem 5

Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be two series in $\mathbb{C}$ converging to A and B respectively. The **Cauchy product** of these two series is defined as the series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that if one of the series $\sum x_n$, $\sum y_n$ converges absolutely, then their Cauchy product converges to AB, this result is known as **Merten's theorem**. Further, consider the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}},$$

which converges by the alternating series test (Leibniz criterion) but not absolutely. Show that the n-th term of its Cauchy product with itself is

$$c_n = \sum_{k=0}^{n} \frac{(-1)^n}{\sqrt{(k+1)(n-k+1)}}$$

and verify that $|c_n| \geq 1$ for all n, so that $c_n \not\to 0$. Conclude that the Cauchy product of two conditionally convergent series need not converge.

Problem 6

Let $\sum a_n (z - z_0)^n$ be a power series with radius of convergence R given by Hadamard's formula. Then show that

$$R = \lim \left| \frac{a_n}{a_{n+1}} \right|$$

provided that this limit exists or equals ∞.

Problem 7

Use Merten's theorem to show that

$$\sum_{k=0}^{\infty} \frac{z^k}{k!} \sum_{k=0}^{\infty} \frac{w^k}{k!} = \sum_{k=0}^{\infty} \frac{(z+w)^k}{k!}.$$

Problem 8

Let (f_n) be a sequence of complex functions on $\Omega \subseteq \mathbb{C}$. Suppose that for every non-negative integer n, there exists $M_n > 0$ such that $|f_n| \leq M_n$ for all $z \in \Omega$ and such that $\sum M_n$ is convergent. Hence show that the series $\sum f_n$ is uniformly convergent throughout. This result is called the **Weierstrass criterion** for complex series of functions.